

Nasij Assumptions Register

This register captures internal modeling assumptions used across the Nasij estimate documents. Values are base-case unless explicitly marked otherwise.

External Baseline Assumptions

ID	Assumption	Base Value	Rationale
A01	Pilot geography coverage in Year 1	3 wilayas	Keeps rollout manageable while covering multiple supply patterns.
A02	Active source nodes (farmers, butchers, third parties) by end of Year 1	420	Operationally feasible onboarding target for a pilot with field teams.
A03	Average declared wool volume per active source per year	170 kg	Conservative blend of small and medium suppliers.
A04	Declaration-to-collection realization ratio	88%	Accounts for cancellations, access failures, and no-shows.
A05	Average gross-to-net reduction during depot prep (D0 to D1)	12%	Covers impurities and manual sorting losses.
A06	Average net reduction during wash stage (D1 to D2)	23%	Reflects moisture and contaminant removal in wash operations.
A07	Transformation conversion efficiency (D2 to D3/D4 usable output)	90%	Includes handling and process loss before commercial output.
A08	Year 1 average traceability service fee	220 DZD/kg processed	Balanced price point for traceability value and early adoption.
A09	Year 1 average logistics fee per pickup mission	2,800 DZD	Baseline pickup fee for route execution and handling at pilot scale.

ID	Assumption	Base Value	Rationale
A10	Year 1 alert handling fee for non-compliance events	900 DZD/event	Cost recovery for exception workflow effort and escalation handling.

Demand, Capacity, and Growth Assumptions

ID	Assumption	Base Value	Rationale
A11	Source-node annual growth rate (Y2 to Y5)	35%	Aggressive but realistic with institutional partnerships.
A12	Average missions per active source per year	3.4	Reflects repeated pickups during seasonal cycles.
A13	Fraction of missions producing discrepancy alerts	14% in Y1, down to 8% by Y5	Better process discipline should reduce avoidable mismatches.
A14	Collection-stage QR tagging adoption	100% from go-live	Mandatory for chain continuity and batch identity.
A15	Depot-to-wash handover digital confirmation rate	97% in Y1, 99% by Y3	Minor operational slippage expected early, then standardization.
A16	Offline data-capture dependency in field missions	45% of missions in Y1	Connectivity variability in rural/remote collection areas.
A17	Offline queue replay success within 24h	94% in Y1, 98% by Y3	Reliability target for sync architecture hardening.
A18	Average transformation lot size	380 kg input equivalent	Supports planning of downstream D3/D4 throughput.
A19	Output split after transformation	62% D3, 38% D4	Initial product mix assumption for commercialization planning.

ID	Assumption	Base Value	Rationale
A20	Contracted buyer retention (annual)	82% in Y1, 89% by Y5	Improved transparency expected to increase buyer confidence.

Cost and Operating Assumptions

ID	Assumption	Base Value	Rationale
A21	Field operations fixed cost (Year 1)	18.6M DZD	Team, vehicles, supervision, and dispatch overhead.
A22	Platform engineering and cloud cost (Year 1)	12.4M DZD	Mobile, backend, dashboard, storage, and observability stack.
A23	Compliance, audit, and governance cost (Year 1)	3.2M DZD	Data protection controls, policy work, and audit process.
A24	Training and change-management cost (Year 1)	2.7M DZD	SOP rollout across collectors, depot, wash, and transform teams.
A25	Variable logistics cost per mission (Year 1)	690 DZD	Fuel, loading/unloading, and consumables per trip.
A26	Variable processing support cost per processed kg	27 DZD/kg	Labels, QA effort, and digital processing overhead.
A27	Annual inflation pass-through on operating costs	+4.0%	Aligned with national inflation context.
A28	Annual fee escalation on service pricing	+3.0%	Slightly below inflation to preserve adoption momentum.
A29	R&D and contingency reserve	6% of annual operating cost	Protects delivery against technical and operational surprises.

ID	Assumption	Base Value	Rationale
A30	Depreciation proxy for equipment and infrastructure	8% of relevant capex envelope	Simplified planning proxy for estimate-level modeling.

Financial and Scenario Rules

ID	Assumption	Base Value	Rationale
A31	Working-capital buffer	4 months of average operating cost	Guards against seasonality and payment delays.
A32	Revenue recognition delay for institutional contracts	45 days average	Reflects common enterprise billing cycles.
A33	Conservative scenario revenue multiplier	0.82x base	Lower adoption + slower route utilization.
A34	Conservative scenario cost multiplier	1.09x base	Lower efficiency with relatively fixed field overhead.
A35	Upside scenario revenue multiplier	1.18x base	Faster onboarding and stronger commercial throughput.
A36	Upside scenario cost multiplier	1.04x base	Scale gains partially offset by growth complexity.
A37	Break-even KPI definition	Annual operating profit ≥ 0	Standard operating break-even threshold.
A38	Minimum target gross margin by Year 5	31%	Indicates healthy service economics at scale.
A39	Risk reserve trigger	Alert rate $> 16\%$ for 2 consecutive quarters	Requires process correction budget release.

ID	Assumption	Base Value	Rationale
A40	Year 5 transformation traceability coverage	>= 96% lots with complete stage linkage	End-state chain visibility objective.

Derived Formula Notes

- Processed gross volume per year = active sources x avg declared volume x realization ratio [A02], [A03], [A04].
- Net washed equivalent = processed gross x (1 - D0/D1 loss) x (1 - D1/D2 loss) [A05], [A06].
- Service revenue base = (processed gross x per-kg fee) + (missions x logistics fee) + (alerts x handling fee) [A08], [A09], [A10], [A12], [A13].
- Scenario outputs apply multipliers in [A33]-[A36].